

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15D.1

CARBOXYLIC ACID DERIVATIVES: ACYL CHLORIDES

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15D.1: Carboxylic Acid Derivatives: Acyl Chlorides
TOTAL QUESTIONS:	3 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	8 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15D.1

Q1. 1. (a) Ethanoyl chloride reacts vigorously with water. Write the equation for this reaction and state the observation.

[2 Marks]**WCH14/01/Jan23/Q8(a)**

Q2. (b) Compare the reactivity of ethanoyl chloride and chlorobenzene towards nucleophilic attack by water. Explain the difference.

[3 Marks]**WCH14/01/Jan23/Q8(b)**

Q3. (c) Write equations for ethanoyl chloride reacting with (i) ethanol and (ii) concentrated ammonia.

[3 Marks]**WCH14/01/Jan23/Q8(c)**

SECTION B: WORKED MARK SCHEME — 15D.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Ethanoyl chloride reacts vigorously with water. Write the equation for this reaction and state the observation. 1. $\text{CH}_3\text{COCl} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{HCl}$. 2. Misty / steamy white fumes of HCl gas observed.	[2]
Q2	Q2: (b) Compare the reactivity of ethanoyl chloride and chlorobenzene towards nucleophilic attack by water. Explain the difference. 1. Ethanoyl chloride is extremely reactive due to δ^+ carbonyl carbon attached to two electronegative atoms (O and Cl). 2. Chlorobenzene is unreactive because p-orbital lone pair on Cl overlaps with π -system of benzene ring. 3. Strengthens C-Cl bond and increases electron density of ring $\rightarrow$ repels nucleophiles.	[3]
Q3	Q3: (c) Write equations for ethanoyl chloride reacting with (i) ethanol and (ii) concentrated ammonia. 1. (i) $\text{CH}_3\text{COCl} + \text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + \text{HCl}$ (ethyl ethanoate + HCl). 2. (ii) $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$ (ethanamide + ammonium chloride).	[3]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15D.1**FAQ #1: Why does ethanoyl chloride react with 2 moles of ammonia?****Category: Stoichiometry & Mechanisms****COMMON EXAMINER TRAP / MISTAKE:**

Writing $\text{CH}_3\text{COCl} + \text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{HCl}$ without accounting for HCl reacting with excess NH_3 .

MODEL ANSWER / EXAMINER EXPECTATION:

The reaction produces HCl gas which immediately reacts with a second mole of basic ammonia to form solid ammonium chloride (NH_4Cl). Thus 2 moles of NH_3 are required per mole of acyl chloride.